Dexterous Hand Manipulation

Learning Methods Survey (2023-2026)

1. Training Methods Comparison

The following table summarizes the main training paradigms for dexterous hand manipulation, with representative works, advantages, and limitations.

Method	Representative	Venue	Pros	Cons	Best For
RL (PPO)	OpenAI, BiDexHands	IJRR19,ICLR25	Contact-rich	Sample hungry	In-hand rot
RL (SAC)	DexGraspRL	2025	More efficient	Still millions	Grasping
Imitation	DexMV, DexUMI	ECCV22,CoRL25	Natural demos	Needs data	Tool use
Diffusion	ALOHA, UniDexFPM	DeepMind24	Multi-modal	Slow ~10Hz	Complex seq
ACT	FTACT, SERNF	2024-25	Fast inference	Less express	Real-time
Teacher-Stu	DextrAH-RGB	2024	Sim->real	Two-stage	Vision policy
World Model	DexWM	2025	Cross-embod	Model error	Video->robot
VLA+low	DexVLG, OpenVLA	ICCV25	Lang+prec	Complex pipe	General
LLM Reward	Eureka, DrEureka	ICLR24,RSS24	Auto reward	LLM cost	Reward design

2. 2024-2025 Dominant Training Pipeline

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Current best practice:

  Step 1: Teacher Policy (RL, PPO/SAC in IsaacGym/Isaac Lab)
    - obs: joint angles + object pose + contact forces (privileged)
    - Domain randomization: physics, actuation, object shape

  Step 2: Student Policy (Distillation via BC / DAgger)
    - Input: RGB / depth / point cloud (real-sensor compatible)
    - Output: joint velocities or positions

  Step 3: Sim-to-Real Transfer
    - Zero-shot (domain randomization)
    - OR few-shot fine-tuning (SERNF/DEFT, <100 real episodes)

  Alternative (rising):
    - Teleoperation (DexUMI / Bunny-VisionPro) -> 50-200 demos
    - Diffusion Policy / ACT on demos -> deploy directly
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3. Teleoperation & Data Collection

DexUMI: Human Hand as Universal Manipulation Interface

CoRL 2025

Human hand + exoskeleton + robot hand inpainting. 86% success across multiple platforms. Bridges embodiment gap via hardware adapter + visual inpainting. Key teleoperation system for dexterous demos.

Bunny-VisionPro: Real-Time Bimanual Dexterous Teleoperation

ICRA 2024

Apple Vision Pro for bimanual teleoperation. Dual 24-DoF system (xArm-7 + Ability hands). Outperforms prior teleop systems.

UMI: Universal Manipulation Interface

ICRA 2024

Hand-held gripper for portable data collection. Zero-shot generalization to novel objects/environments. Variants: MV-UMI, DexUMI, exUMI.

DexCap: Scalable Portable Mocap Data Collection

RSS 2024

EMF tracking gloves + RGB-D LiDAR. 3x faster than teleoperation. 72% success from 30-minute mocap sessions. Backpack-mounted.

AnyTeleop: General Vision-Based Dexterous Teleoperation

ICRA 2023

Hardware-agnostic. Only needs RGB/RGB-D camera. Higher success than specialized systems in 8/10 tasks. Works over internet.

4. RL & Policy Learning

BiDexHands: Human-Level Bimanual Dexterous Manipulation

ICLR 2025

Dual Shadow Hands. 50+ bimanual tasks. IPPO teacher -> vision student. Isaac Gym. Most widely used dexterous RL benchmark.

Eureka: Human-Level Reward Design via LLM

ICLR 2024

GPT-4 auto-generates RL reward functions. Outperforms human-designed rewards on 83% of 29 tasks. 52% average improvement. Shadow Hand pen spinning tricks. Eliminates manual reward engineering.

DrEureka: LLM-Guided Sim-to-Real Transfer

RSS 2024

LLM automates domain randomization. Eliminates manual DR tuning. Validated on quadruped + dexterous manipulation. Builds on Eureka.

DextrAH-RGB: Visuomotor Policies to Grasp Anything

2024

Teacher (state RL, IPPO) -> Student (vision, DAgger). Isaac Sim. Zero-shot sim-to-real. Successful diverse object grasping.

SAPO: Stabilizing RL in Differentiable Multiphysics Sim

2024

Short-horizon autodiff (10-50 steps) + long-horizon RL (SAC). Addresses gradient exploding/vanishing. Most relevant to our P(F) differentiable sim approach.

5. Diffusion & Imitation Learning

ALOHA Unleashed: Simple Recipe for Robot Dexterity

Google Dee

Diffusion Policy on 26k+ demos. Shoelace tying, shirt hanging, gear insertion. Key insight: large-scale data + diffusion = complex dexterity. $32K bimanual hardware. Open-source.

DexMV: Learning from Human Videos

ECCV 2022

YouTube videos -> hand pose extraction -> robot demos. 83% success on real Allegro Hand. Eliminates need for VR/teleoperation.

UniDexFPM: Dexterous Functional Pre-Grasp Manipulation

2024

Mixture-of-experts -> diffusion policy. 72.6% across 30+ object categories. Isaac Gym. Pre-grasp reorientation before grasping.

Tilde: Diffusion + DAgger for Dexterous Manipulation

RSS 2024

Diffusion policy + human DAgger corrections. DeltaHand (12-DoF). 90% success across 7 tasks. Real-time expert corrections.

DexDeform: Dexterous Deformable Object Manipulation

ICLR 2023

Human demos + differentiable physics optimization. 6 deformable tasks (Play-Doh). Most relevant for deformable + dexterous combination.

6. World Models & Foundation Models

DexWM: Dexterous Interaction World Model

2025

900+ hours human and robot data. Cross-embodiment transfer. Beats Diffusion Policy by 50%+ on Franka + Allegro. Zero-shot transfer.

EgoDex: Learning from Large-Scale Egocentric Video

Apple 2025

829 hours egocentric video. 194 household tasks. Apple Vision Pro capture. Largest dexterous manipulation dataset. Open-source.

HumanPlus: Humanoid Shadowing and Imitation

CoRL 2024

33-DoF humanoid with dual 6-DoF hands. Shoe wearing, warehouse unloading, typing. 60-100% success from ~40 demos.

OpenVLA: Open-Source Vision-Language-Action Model

2024

7B VLA. 970k real-world demos. 50%+ success. Fully open-source.

RoboPianist: Dexterous Piano Playing

CoRL 2023

Learns 150 piano pieces. Deep RL with bimanual anthropomorphic hands. Demonstrates extreme coordination. RP1M: 1M expert trajectories.

7. Hardware Platforms

Hand	Cost	DOF	Notes
Shadow Hand	$100K+	24	RL research standard. UK ARIA $57M. OpenAI, DeepMind
Allegro Hand	$16K	16	DexMV, DexPoint, DexCap. Good middle ground
LEAP Hand	$2K	16	RSS 2024. 3D-printed. Outperforms Allegro. Open-source
RUKA v2	<$2K	15	NYU 2025. Tendon-driven. Fishing line. Open-source
ORCA Hand	$1.5-6K	17	ETH 2025. 10K+ cycles. Optional tactile. MIT license
Ability Hand	$20K+	10	Bunny-VisionPro. Commercial grade
DeltaHand	Research	12	CMU. Non-anthropomorphic delta robot design
DexRobot 021	Commercial	19	CES 2026. 23 sensors. Rubik's cube solving

8. Surgical Dexterous Manipulation

Dexterity-Enhancing Handheld Robot for Endoscopic Neurosurgery

npj Robotic

Panda Surgical handheld robot. Interchangeable end-effectors (forceps, curettes, dissectors). 2-3 DoF per tool. First clinical validation for dexterous neurosurgery. Bimanual without brain compression.

SutureBot: Autonomous End-to-End Surgical Suturing

2025 (arXiv

da Vinci Research Kit. 1,890 demos. Needle pickup, tissue insertion, knot tying. Imitation learning. First comprehensive suturing framework.

Dex Surgical: Commercial Surgical Robotics

Commercia

PrecisionDex 3D technology. 7-DoF dexterous laparoscopic tools. Minimally invasive surgery. Commercial market solution.

Gap: No existing work combines dexterous hand + CUSA tool + differentiable debulking simulation. This is our unique opportunity.

9. Application: Dexterous CUSA Debulking

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Proposed System: Dexterous Hand + CUSA Tool + P(F) Differentiable Sim

Hardware:
  - LEAP Hand ($2K) or Allegro Hand ($16K)
  - CUSA-like tool mounted on fingertips
  - RealSense depth camera for observation

Training Pipeline:
  Step 1: Teacher Policy (RL, PPO/SAC)
    - Sim: Isaac Lab (hand) + Warp (P(F) tissue)
    - obs: hand joints + CUSA tip + local P(F) distribution
    - action: hand joint velocities
    - reward: +volume_removed - lambda*nerve_damage(P(F))

  Step 2: Student Policy (vision-based)
    - Teacher -> Student via DAgger
    - OR Diffusion Policy on teleop demos

  Step 3: Sim-to-Real
    - Domain randomization on tissue properties
    - Short-horizon autodiff for local refinement
    - Transfer to LEAP Hand + silicone phantom

Paper Framing:
  "Differentiable Geometry-Aware Debulking with Dexterous Manipulation:
   P(F)-guided RL for Vestibular Schwannoma Surgery"

Target Venues: CoRL, ICRA, RSS
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10. Recommended Reading Priority

Paper	Venue	Why Read
SAPO (RL + DiffSim)	2024	Our exact approach: short diff + long RL
DexUMI	CoRL 2025	Best teleop for dexterous data collection
BiDexHands	ICLR 2025	PPO benchmark for dexterous RL
ALOHA Unleashed	DeepMind 2024	Diffusion policy for complex dexterity
Eureka / DrEureka	ICLR24/RSS24	LLM auto reward + domain randomization
DextrAH-RGB	2024	Teacher-student sim-to-real pipeline
LEAP Hand	RSS 2024	Low-cost hardware ($2K). Open-source
DexWM	2025	Cross-embodiment world model from video
DexDeform	ICLR 2023	Dexterous + deformable + diff physics
Neurosurgery Robot	npj 2025	First clinical dexterous neurosurgery
DexMV	ECCV 2022	YouTube video -> robot policy
EgoDex	Apple 2025	829hr dataset. Largest dexterous dataset